

The background is a light blue gradient. A large white rectangular box is centered, containing the title and authors. Surrounding the box are various medical icons: a stethoscope with a black tube and silver chest piece on the left; several syringes and pills in various colors (red, blue, yellow, orange) scattered around the top and bottom edges; and some faint, light blue diagonal lines in the corners.

Surgery Timing

BIS 15L Winter Quarter 2026 Group 2

By: Ale, Daniella, Aaliyah

Dataset Intro

Who: 32,001 elective general surgery patients

Where: Cleveland Clinic

When: between 2005 and 2010.

Why:

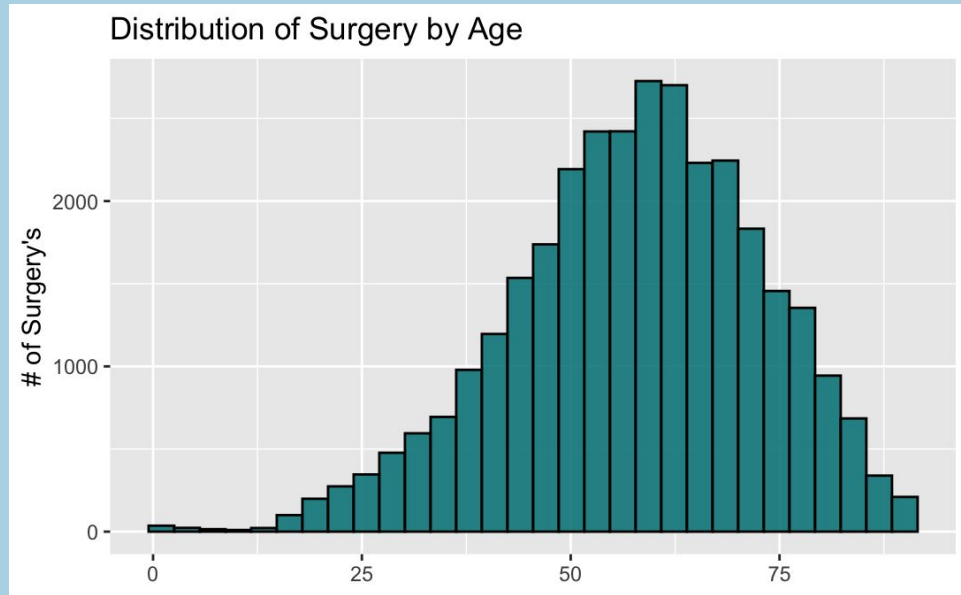
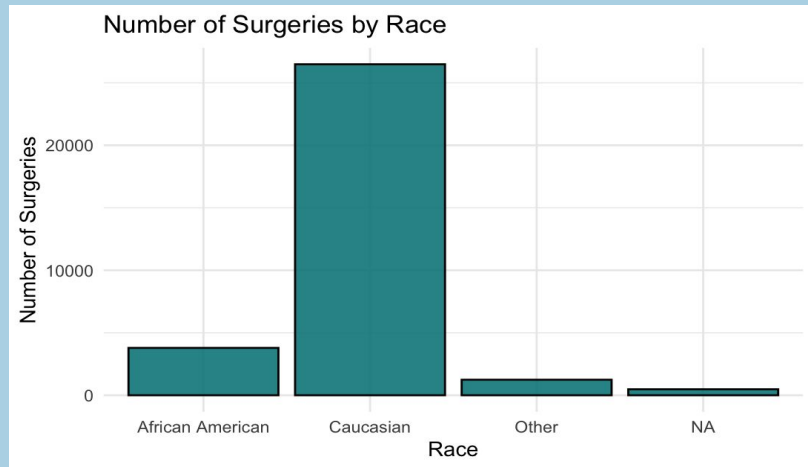
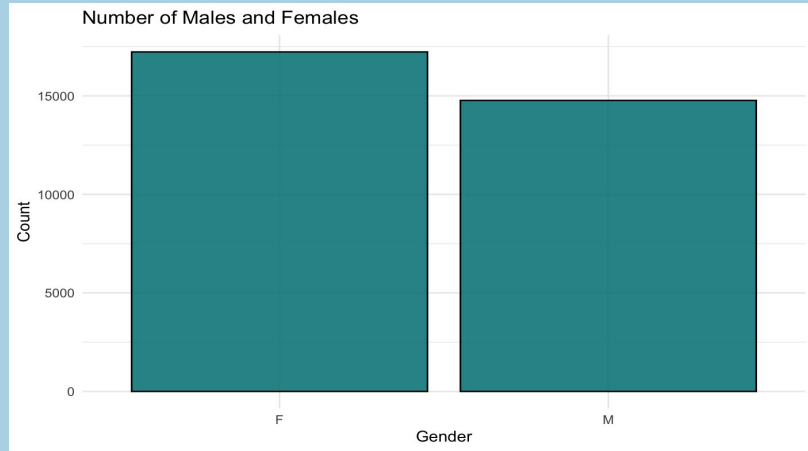
- Lack of sleep and heavy workloads may affect surgical performance.
- Determine whether the time of surgery impacts patient outcomes.

Our Goal

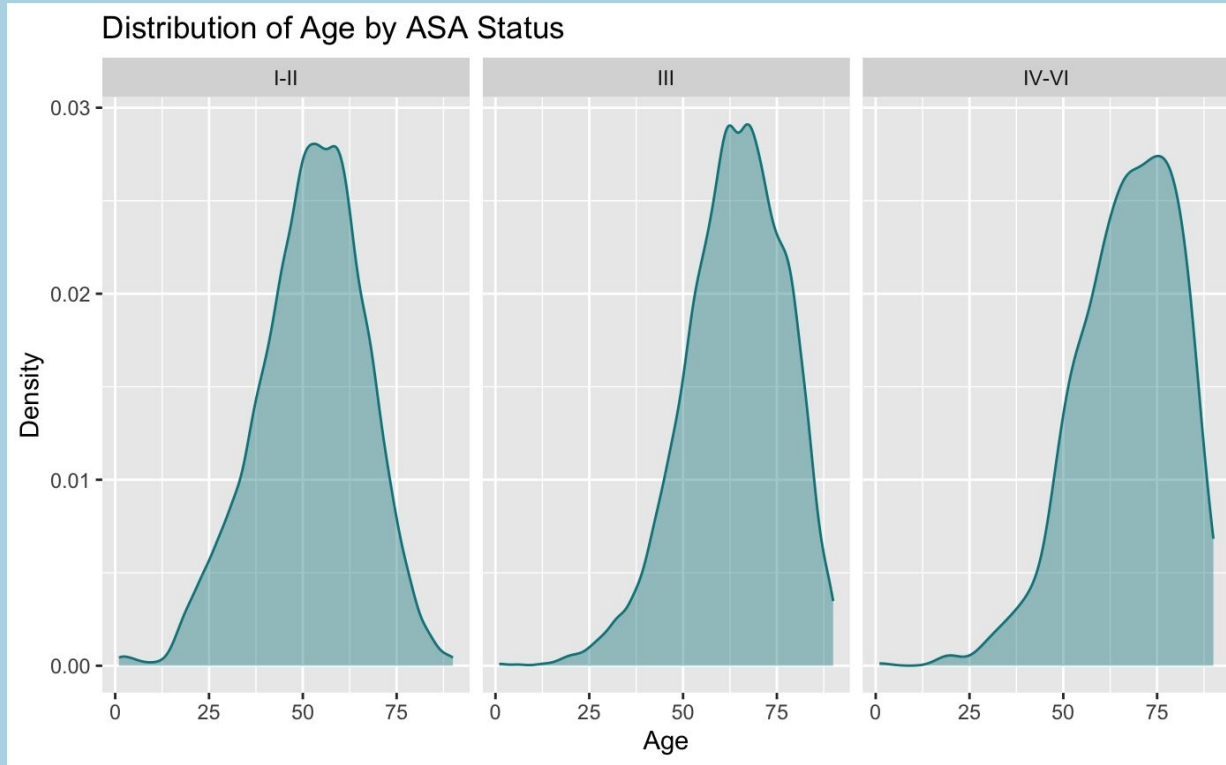
Analyze the dataset to determine whether surgery timing (hour, day, month, or moon phase) is associated with patient mortality.



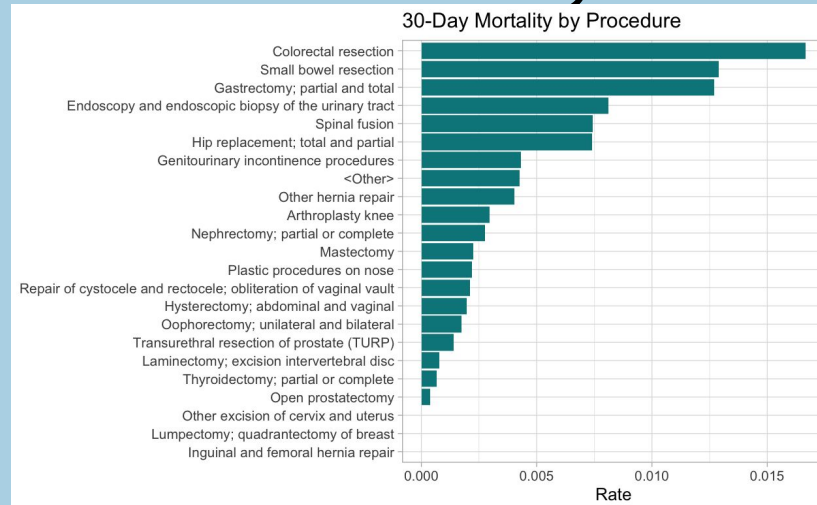
Patient Diversity: Demographic



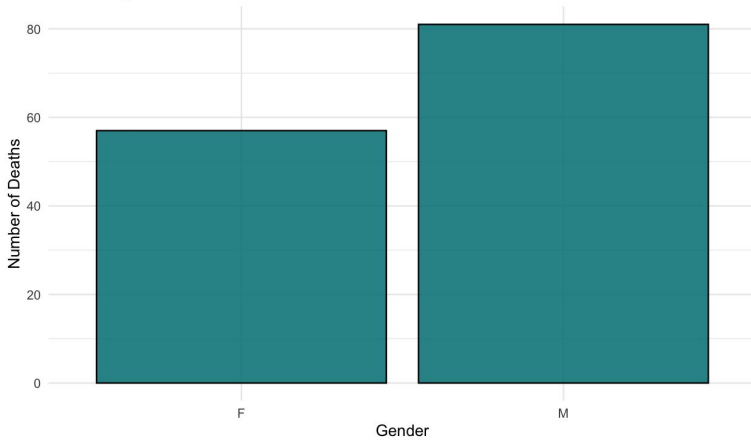
Patient Diversity: Health



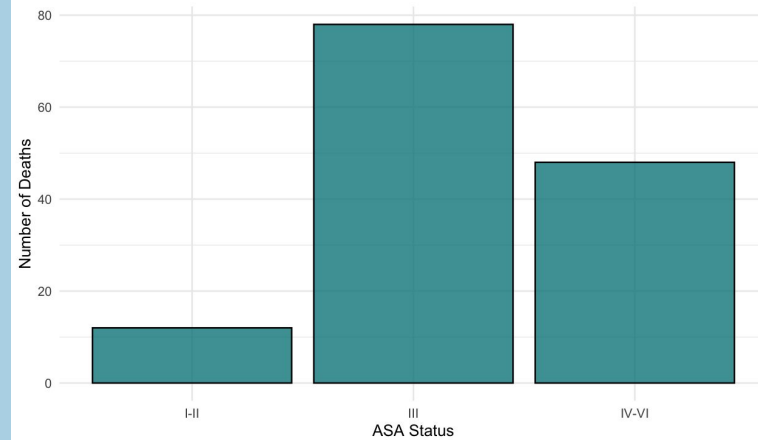
Mortality



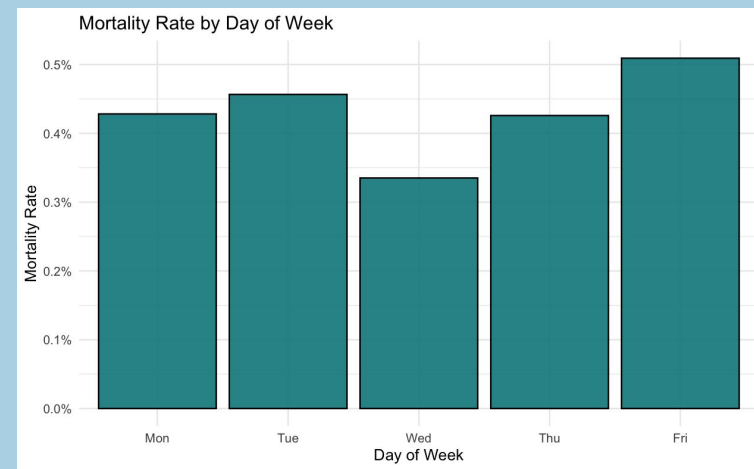
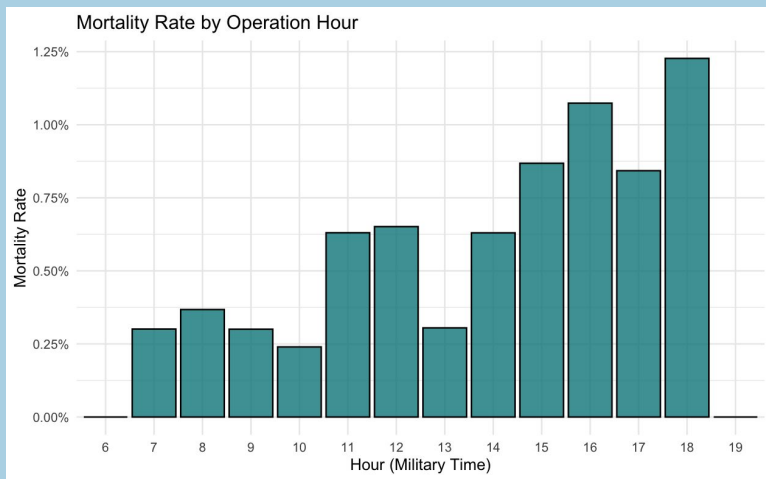
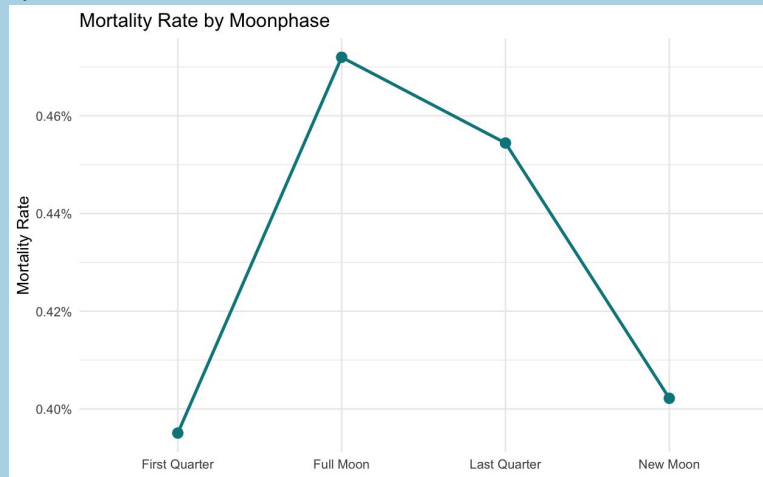
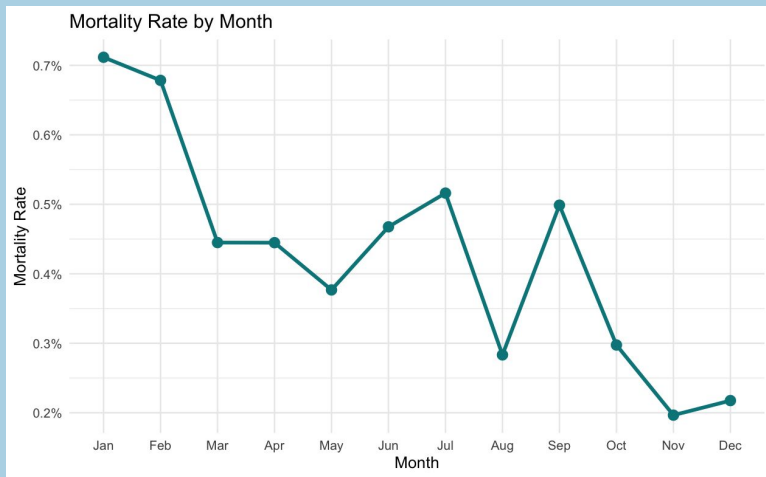
Deaths by Gender



Death Count Based on ASA Status

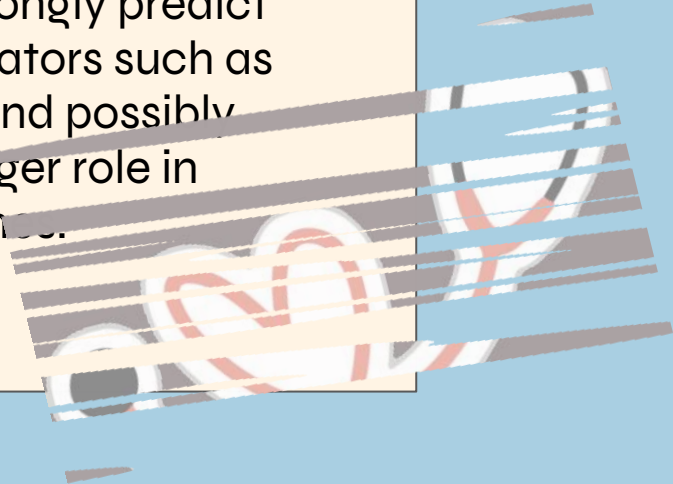


Shiny App



Conclusion

The goal of this project was to analyze whether the timing of surgery is associated with patient outcomes. While factors such as month, hour of surgery, and moon phase were explored, the results suggest that timing alone does not strongly predict mortality. Instead, patient health indicators such as ASA status, existing complications, and possibly demographics appear to play a larger role in determining surgical outcomes.



The background is a solid light blue. A large white rectangle is centered, containing the text 'THANK YOU!'. Surrounding this rectangle are various medical icons: a stethoscope with a black tube and silver chest piece is positioned on the left, with its earbuds pointing towards the top left. Several syringes and pills are scattered around the rectangle. Pills come in various colors: red and blue, yellow and pink, and orange and white. Some pills are whole, while others are broken. The overall theme is medical or healthcare.

THANK YOU!